Computer Vision

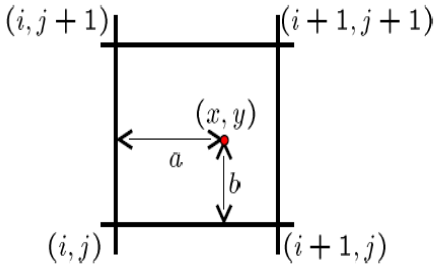
EMARO- *European Master on Advanced Robotics*
Robotics Engineering Master Degree

Lab Session n. 1

Image warping and bilinear interpolation

Given a coordinate transform *(x,y)=h(x',y')* and a source image *f(x',y')*, compute a transformed image *g(x,y) = f(h(x',y'))*.

- Perform backward warping (from the output image to the input image) with a bilinear interpolation, see Fig. 1 (it is also possible to use the function `griddata`).
- Example of transformations, see Fig. 2: 1) translation and rotation (try different angles). 2) To use the data of the file *data.mat* to warp the pixels of the test images.



$$
\begin{aligned}
f(x,y) = \; & (1-a)(1-b) \quad f[i,j] \\
& +a(1-b) \quad f[i+1,j] \\
& +ab \quad f[i+1,j+1] \\
& +(1-a)b \quad f[i,j+1]
\end{aligned}
$$

Fig.1 Bilinear interpolation

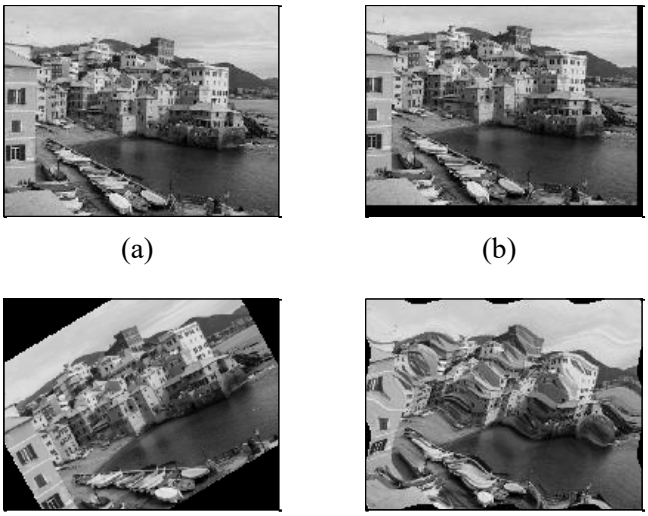


Fig.2 (a) Test image (gray scale). (b) Translation. (c) Rotation. (d) Transformation by using data.mat.

Notes:

- About the code (M-files):
  - You have to use relative paths.
  - You have to write and use functions
  - You have to provide a script to test your code.